

## KINSELLA, AB, Canada

WMO: 718885

Lat: 53.0469N

Lon: 111.5473W

Elev: 2300

StdP: 13.51

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -26.7      | -19.8 | -32.8                         | 0.9 | -26.5 | -25.7 | 1.4 | -19.7 | 23.8                  | 22.3 | 20.9 | 21.5 | 3.7                   | 280  | 0.688 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.2                   | 83.2            | 62.7 | 80.0 | 61.7 | 77.0 | 60.8 | 66.6                | 77.7 | 64.6 | 75.6 | 62.8 | 73.1 | 9.5                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.6                            | 92.5 | 72.5 | 60.4 | 85.4 | 69.5 | 58.6 | 80.1 | 67.3 | 32.6          | 78.3 | 30.9 | 76.0 | 29.5 | 73.3 | 74.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 23.0              | 20.1 | 17.9 | -37.4                      | 89.0 | 6.1                | 3.8 | -41.7                                              | 91.7 | -45.3      | 93.9 | -48.7      | 96.0 | -53.2      | 98.7 |
|                   |      |      | -37.5                      | 71.4 | 6.1                | 2.3 | -41.9                                              | 73.0 | -45.5      | 74.3 | -48.9      | 75.6 | -53.3      | 77.2 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 36.3   | 12.1  | 9.1   | 22.0  | 36.9  | 51.2  | 58.0  | 62.4  | 60.6  | 51.5  | 38.2  | 20.9  | 11.2  |
|                                                                        | DBStd       | 22.37  | 15.07 | 15.83 | 14.56 | 9.67  | 7.74  | 5.48  | 4.71  | 5.55  | 8.37  | 8.65  | 13.21 | 13.73 |
|                                                                        | HDD50       | 6208   | 1175  | 1145  | 869   | 402   | 79    | 7     | 0     | 2     | 80    | 373   | 874   | 1202  |
|                                                                        | HDD65       | 10523  | 1640  | 1565  | 1334  | 844   | 430   | 218   | 109   | 154   | 408   | 830   | 1324  | 1667  |
|                                                                        | CDD50       | 1220   | 0     | 0     | 0     | 8     | 115   | 248   | 384   | 331   | 126   | 8     | 0     | 0     |
|                                                                        | CDD65       | 60     | 0     | 0     | 0     | 0     | 2     | 9     | 28    | 18    | 3     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1379   | 0     | 0     | 0     | 6     | 130   | 164   | 436   | 425   | 216   | 2     | 0     | 0     |
|                                                                        | CDH80       | 274    | 0     | 0     | 0     | 0     | 15    | 25    | 88    | 87    | 59    | 0     | 0     | 0     |
|                                                                        | WSAvg       | 8.0    | 8.0   | 7.8   | 8.3   | 9.3   | 8.8   | 8.4   | 7.7   | 6.8   | 7.6   | 8.4   | 7.5   | 7.4   |
|                                                                        | PrecAvg     | 16.1   | 0.9   | 0.5   | 0.6   | 1.2   | 1.6   | 2.8   | 2.9   | 2.3   | 1.3   | 0.7   | 0.7   | 0.7   |
| (15) Precipitation                                                     | PrecMax     | 23.1   | 2.8   | 0.9   | 1.3   | 3.7   | 2.6   | 4.9   | 5.9   | 4.9   | 3.1   | 1.3   | 3.3   | 1.3   |
|                                                                        | PrecMin     | 8.5    | 0.2   | 0.0   | 0.2   | 0.2   | 0.2   | 0.6   | 1.1   | 0.4   | 0.1   | 0.0   | 0.1   | 0.2   |
|                                                                        | PrecStd     | 3.3    | 0.6   | 0.3   | 0.3   | 0.8   | 0.6   | 1.1   | 1.3   | 1.0   | 0.9   | 0.4   | 0.7   | 0.4   |
|                                                                        | PrecStd     | 3.3    | 0.6   | 0.3   | 0.3   | 0.8   | 0.6   | 1.1   | 1.3   | 1.0   | 0.9   | 0.4   | 0.7   | 0.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 43.9  | 43.5  | 55.0  | 71.6  | 82.1  | 83.1  | 86.2  | 85.8  | 86.8  | 69.2  | 55.5  | 42.3  |
|                                                                        |             | MCWB   | 37.4  | 37.8  | 44.8  | 49.5  | 56.4  | 61.0  | 67.7  | 63.0  | 59.3  | 53.7  | 44.3  | 36.1  |
|                                                                        | 2%          | DB     | 40.1  | 38.1  | 48.7  | 64.5  | 78.0  | 78.7  | 82.6  | 82.0  | 80.3  | 61.6  | 46.6  | 38.1  |
|                                                                        |             | MCWB   | 35.3  | 34.0  | 41.0  | 46.2  | 55.0  | 60.3  | 65.6  | 63.7  | 57.6  | 48.3  | 39.9  | 33.6  |
|                                                                        | 5%          | DB     | 37.3  | 35.5  | 43.2  | 58.6  | 73.3  | 74.7  | 78.8  | 78.8  | 74.5  | 57.4  | 41.4  | 34.2  |
|                                                                        |             | MCWB   | 33.6  | 32.2  | 37.1  | 43.7  | 53.6  | 59.4  | 64.4  | 62.7  | 56.2  | 46.1  | 36.8  | 30.3  |
|                                                                        | 10%         | DB     | 33.6  | 32.5  | 39.3  | 53.8  | 68.9  | 71.6  | 75.6  | 75.3  | 68.8  | 52.6  | 37.4  | 30.6  |
|                                                                        |             | MCWB   | 30.5  | 30.1  | 34.7  | 41.6  | 51.4  | 57.3  | 63.1  | 61.5  | 53.6  | 43.8  | 34.2  | 27.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 37.8  | 37.6  | 45.5  | 50.4  | 58.8  | 64.8  | 70.9  | 69.2  | 61.9  | 54.0  | 45.7  | 36.5  |
|                                                                        |             | MCDB   | 42.9  | 42.7  | 53.4  | 70.5  | 77.9  | 76.5  | 81.5  | 78.8  | 78.0  | 66.5  | 55.1  | 40.9  |
|                                                                        | 2%          | WB     | 35.7  | 34.6  | 40.4  | 46.7  | 56.4  | 62.8  | 67.6  | 66.2  | 59.2  | 49.5  | 40.4  | 33.7  |
|                                                                        |             | MCDB   | 39.9  | 37.5  | 47.8  | 62.1  | 72.8  | 74.2  | 78.5  | 77.1  | 76.3  | 57.7  | 45.8  | 37.9  |
|                                                                        | 5%          | WB     | 33.5  | 32.4  | 37.5  | 44.6  | 54.5  | 60.9  | 65.5  | 64.0  | 57.2  | 47.2  | 37.1  | 30.6  |
|                                                                        |             | MCDB   | 36.9  | 35.0  | 43.0  | 57.2  | 70.0  | 71.7  | 76.6  | 75.3  | 73.1  | 55.7  | 41.0  | 34.0  |
|                                                                        | 10%         | WB     | 30.5  | 29.9  | 34.9  | 42.3  | 52.8  | 59.1  | 63.7  | 62.1  | 55.0  | 44.7  | 34.4  | 27.5  |
|                                                                        |             | MCDB   | 33.4  | 32.8  | 38.8  | 52.5  | 66.8  | 68.5  | 73.8  | 72.7  | 67.1  | 52.5  | 37.3  | 30.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.0  | 20.1  | 18.3  | 21.3  | 25.8  | 22.3  | 22.2  | 25.0  | 25.6  | 21.1  | 16.7  | 16.8  |
|                                                                        |             | MCDBR  | 19.2  | 20.9  | 21.7  | 31.1  | 33.5  | 27.5  | 28.4  | 31.9  | 37.4  | 29.5  | 21.2  | 20.0  |
|                                                                        |             | MCWBR  | 15.6  | 17.4  | 14.9  | 16.1  | 14.4  | 12.6  | 14.6  | 15.2  | 17.3  | 17.2  | 14.7  | 16.3  |
|                                                                        |             | MCDDBR | 19.5  | 20.0  | 21.1  | 28.9  | 29.8  | 23.4  | 26.1  | 27.3  | 33.6  | 26.4  | 19.9  | 19.6  |
|                                                                        | 5% WB       | MCWBR  | 16.2  | 17.0  | 14.8  | 15.5  | 13.8  | 12.0  | 14.2  | 14.3  | 16.4  | 15.9  | 14.2  | 16.5  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.255 | 0.286 | 0.297 | 0.330 | 0.350 | 0.366 | 0.361 | 0.374 | 0.314 | 0.278 | 0.253 | 0.231 |
|                                                                        | taud        |        | 2.306 | 2.260 | 2.366 | 2.362 | 2.355 | 2.343 | 2.381 | 2.323 | 2.465 | 2.540 | 2.448 | 2.352 |
|                                                                        | Ebn at Noon |        | 236   | 260   | 285   | 287   | 285   | 280   | 280   | 268   | 276   | 267   | 241   | 229   |
|                                                                        | Edh at Noon |        | 18    | 27    | 30    | 34    | 36    | 37    | 35    | 36    | 27    | 20    | 16    | 15    |
|                                                                        | RadAvg      |        | 328   | 662   | 1134  | 1496  | 1842  | 1891  | 1974  | 1608  | 1133  | 665   | 328   | 244   |
| (45) All-Sky Solar Radiation                                           | RadStd      |        | 32    | 49    | 87    | 136   | 127   | 134   | 103   | 124   | 121   | 91    | 34    | 23    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (77 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page